KAIZEN MATHS

Lesson starter

Algebraic substitution

Multiple variables - mixed practice

4 questions · about 5 minutes

Name _____

Evaluate each expression. Use brackets for negative values.

S1 Given $x = 6, y = 5$

$$6x + 5y$$

S2 Given $a = -2, b = -1, c = 0$

$$4a - 4b + 3c$$

S3 Given $a = 2, b = -3$

$$a^2 - 6ab + b^2$$

S4 Given $x = 2, y = 1$

$$5x + 2y$$

Show your working

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AS-10 · S · Starter

Show your working

KAIZEN MATHS

Exit ticket

Algebraic substitution

Multiple variables - mixed practice

Name _____

Evaluate each expression.

E1 Given $x = 2, y = 1$

$5x + 2y$

E2 Given $a = -3, b = -4, c = 1$

$2a - 6b + 2c$

E3 Given $a = -3, b = 1$

$a^2 - 6ab + b^2$

AS-10 · E · 3 questions

Show your working

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Exit ticket

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Multiple variables - mixed practice

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Evaluate each expression.

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AS-10 · E · 3 questions

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$5x + 2y$

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AS-10 · E · 3 questions

Show your working

KAIZEN MATHS

Teacher answers & working

AS-10 · Multiple variables - mixed practice

Page 1: two starters. Page 2: four exit tickets. Cut along the dashed lines.

Print student pages single-sided at 100%. Match S and E labels to the solutions below.

S1

Given x=6, y=5

$$6x + 5y$$

Substitute x=6, y=5.

$$6\times 6 + 5\times 5$$

$$36 + 25$$

$$61$$

S2

Given a=-2, b=-1, c=0

$$4a - 4b + 3c$$

Substitute values.

$$4\times(-2) - 4\times(-1) + 3\times 0$$

$$-8 + 4$$

$$-4$$

S3

Given a=2, b=-3

$$a^2 - 6ab + b^2$$

Substitute a=2, b=-3.

$$(2)^2 - 6(2)(-3) + (-3)^2$$

$$4 + 36 + 9$$

$$49$$

S4 / E1

Given x=2, y=1

$$5x + 2y$$

Substitute x=2, y=1.

$$5\times 2 + 2\times 1$$

$$10 + 2$$

$$12$$

E2

Given a=-3, b=-4, c=1

$$2a - 6b + 2c$$

Substitute values.

$$2\times(-3) - 6\times(-4) + 2\times 1$$

$$-6 + 24 + 2$$

$$20$$

E3

Given a=-3, b=1

$$a^2 - 6ab + b^2$$

Substitute a=-3, b=1.

$$(-3)^2 - 6(-3)(1) + (1)^2$$

$$9 + 18 + 1$$

$$28$$